# Study of Pressure Measurement and Calibration of Pressure Gauge Using Dead Weight Tester

## Introduction

A Pressure is the amount of force applied perpendicular to the surface of an object per unit area [1].

$$P = \frac{F}{A}$$

Pressure is generally classified by the reference point used to measure it:

**Absolute Pressure ($\boldsymbol{P_{abs}}$):** When pressure is measured with reference to absolute vacuum pressure then it is called absolute pressure. At this pressure, the molecular momentum is zero [2]. A full vacuum has an absolute pressure reading of 0 PSIA and the average barometric pressure at sea level is $\sim 14.7\ PSIA$ [3]. Absolute pressure is independent from location or altitude, providing a constant, precise measurement.

$$P_{abs} = P_{gauge} + P_{atmospheric}$$

**Gauge Pressure ($\boldsymbol{P_{gauge}}$):** Gauge pressure is the difference between the absolute pressure and the local atmospheric pressure. Gauge pressures can be positive or negative [4]. It is the most commonly used type in industry

$$P_{gauge} = P_{abs} - P_{atmospheric}$$

**Vacuum Pressure:** Vacuum pressure is the pressure in a sealed space that is lower than the surrounding atmospheric pressure, measured as a negative gauge pressure or as a partial vacuum (relative to absolute zero) [5].

$$P_{vac} = P_{atmospheric} - P_{abs}$$

**Atmospheric pressure ($\boldsymbol{P_{atmospheric}}$):** Atmospheric pressure, also known as barometric pressure is the pressure exerted by the column of air above any location on Earth. At sea level, the average value is one atmosphere and that decreases gradually with altitude [6].

**Pressure Variation for Incompressible Fluid:** In an incompressible fluid (a fluid whose density remains constant), pressure varies mainly due to changes in depth or elevation. According to fluid statics, pressure increases as we move deeper into the fluid because of the weight of the fluid above. This relationship is expressed by-

$$P = \rho g h$$

Thus, in an incompressible fluid, pressure depends only on depth and not on the shape or volume of the container.

**Bourdon tube:** A Bourdon tube is a crucial, generally C-shaped, flattened metal tube used in mechanical pressure gauges to measure liquid or gas pressure. When pressurized, the elastic, sealed tube attempts to straighten, and this movement is mechanically transferred to a pointer on a dial [7].

**Diaphragm:** A diaphragm is a flexible, accordion-like, airtight seal constructed from welded metal or plastic diaphragms that converts pressure into movement or separates different pressure zones. It acts as a leak-proof barrier, often in vacuum systems, aerospace, or pumps, offering high precision, durability, and sealing [8].

**Piezoresistive Strain Gauge:** Piezoresistive strain gauges are a semiconductor material which changes in resistance when the material stretched or compressed [9].

**Capacitive Pressure Sensor:** A capacitive pressure sensor is a type of sensor that measures pressure through changes in capacitance.

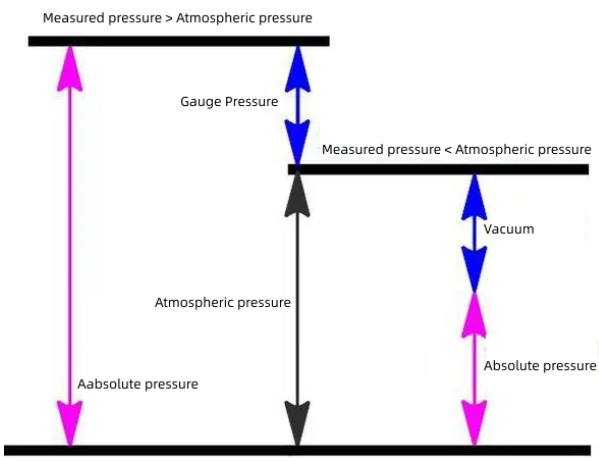
## Objective

- To understand the fundamental concept of pressure and its measurement.
- To study the working principle of pressure gauges, dead weight testers, and pressure sensors.
- To analyze the behavior of pressure sensors under varying pressure conditions.
- To establish the relationship between applied pressure and sensor output through experimental data.

## Working Principle

### Pressure Types

Pressure is defined as the force exerted per unit area and can be expressed in different forms depending on the reference point. Absolute pressure is measured with respect to a perfect vacuum, whereas atmospheric pressure is the pressure exerted by the surrounding air. Gauge pressure represents the difference between the measured pressure and atmospheric pressure. When the measured pressure is greater than atmospheric pressure, it is termed positive gauge pressure, and when it is lower, it is referred to as vacuum pressure. Therefore, absolute pressure is equal to the sum of atmospheric



and gauge pressure, while vacuum pressure represents the difference between atmospheric and absolute pressure.

**Piezometer Tube**

A piezometer tube is one of the simplest devices used to measure pressure. It consists of a vertical transparent tube connected to the point where pressure is to be measured. The liquid rises in the tube to a height proportional to the pressure at that point. The pressure is determined by measuring the height of the liquid column, making it suitable for measuring low pressures in liquids.

**U-Tube Manometer**

A U-tube manometer operates on the principle of hydrostatic equilibrium. It consists of a U-shaped tube partially filled with a manometric fluid such as mercury or water. When pressure is applied at one end, the fluid levels in the two arms of the tube differ. The difference in height between the two columns is proportional to the pressure difference, allowing accurate measurement of gauge pressure.

**Bourdon Tube Pressure Gauge**

The Bourdon tube pressure gauge operates on the principle of elastic deformation. It consists of a curved metallic tube with an oval cross-section. When pressure is applied inside the tube, it tends to straighten due to internal stress. This deformation is transmitted through a mechanical linkage to a pointer, which moves over a calibrated scale to indicate the pressure value.

**Diaphragm Pressure Sensor**

A diaphragm pressure sensor works by utilizing a thin, flexible membrane that deforms when subjected to pressure. The deflection of the diaphragm is directly proportional to the applied pressure. This displacement can be measured either mechanically or converted into an electrical signal, allowing accurate determination of pressure.

**Piezoresistive Strain Gauge Sensor**

The piezoresistive strain gauge sensor is based on the piezoresistive effect, where the electrical resistance of a material changes when it is subjected to mechanical strain. When pressure is applied, it causes deformation in the sensing element, typically made of silicon. This deformation results in a change in resistance, which is measured and correlated to the applied pressure.

**Capacitive Pressure Sensor**

A capacitive pressure sensor operates on the principle of variation in capacitance. It consists of two conductive plates, one fixed and the other flexible. When pressure is applied, the flexible plate deflects, altering the distance between the plates. This change

in separation causes a variation in capacitance, which is measured and used to determine the pressure.

## Result

In this experiment, no direct measurements or experimental data were collected as the lab was primarily theory-based. However, a clear understanding of different types of pressure, including absolute, gauge, and vacuum pressure, was achieved. The operating principles of various pressure measuring devices such as the piezometer tube, U-tube manometer, Bourdon tube, diaphragm sensor, piezoresistive strain gauge, and capacitive pressure sensor were studied and understood. The relationship between applied pressure and corresponding mechanical or electrical response in these devices was also comprehended.

## Discussion

The lab provided a conceptual understanding of pressure measurement techniques and sensor behavior without performing any hands-on experiment. Devices like the piezometer and U-tube manometer rely on hydrostatic principles, making them simple and accurate for low-pressure measurements. In contrast, instruments such as the Bourdon tube and diaphragm sensors utilize mechanical deformation to indicate pressure, which is suitable for a wider range of applications.

Modern sensors, including piezoresistive and capacitive types, convert pressure into electrical signals, offering higher sensitivity and suitability for digital systems. Although no experimental calibration was performed using a dead weight tester, the theoretical concept of calibration and error analysis was discussed. The absence of practical work limited the opportunity to observe real-time deviations and instrument accuracy, but the viva session helped reinforce the theoretical knowledge and clarify concepts.

## Conclusion

The lab successfully provided a theoretical understanding of pressure, its types, and various pressure measurement techniques. Even though no practical experiment was conducted, the study enhanced knowledge about the working principles and applications of different pressure measuring devices and sensors. The viva session further strengthened the understanding of key concepts, making the objectives of the lab effectively achieved.

## Reference

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